

ESSHAN KHATRI

Apprentice – Generative AI & Automation | Python, SQL, LLM Integration, RAG & Workflow Automation
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PROFESSIONAL SUMMARY

AI & Machine Learning undergraduate with a strong foundation in Python, SQL, and Object-Oriented Programming, applied to building data pipelines, automating reporting, and validating data accuracy. Hands-on project experience designing a Retrieval-Augmented Generation (RAG) pipeline with LangChain and Hugging Face Transformers — covering web ingestion, chunking, vector embeddings, similarity search, LLM integration, and prompt design/testing — with a habit of reviewing AI-generated outputs to reduce hallucination before use. Practical experience training and evaluating ML models with Scikit-learn and PyTorch. Comfortable breaking problems into step-wise, automatable components that combine rule-based logic with GenAI, and building foundational knowledge of Agentic AI and LLM orchestration concepts. Six completed certifications spanning Generative AI, Python, and data analytics. Looking for an Apprentice role on a GenAI & Automation team to contribute to and learn from real-world automation, analytics, and reporting workflows.

TECHNICAL SKILLS

Languages & Data: Python (Pandas, NumPy), SQL (Joins, Aggregations, Filtering), Object-Oriented Programming (OOP), Data Structures & Algorithms, C++

Generative AI & LLM: RAG (Retrieval-Augmented Generation), LangChain, Hugging Face Transformers, Prompt Engineering / Prompt Design & Testing, LLM Integration, Vector Embeddings & Similarity Search, Context-Grounded Generation, Agentic AI (foundational concepts), Text Classification & Extraction

Machine Learning: Scikit-learn, Model Training & Evaluation, PyTorch, TensorFlow

Backend & Database: REST APIs & Tool Integration, Flask, MySQL, SQLite, Supabase, Database Management, Model Deployment

Automation & Reporting: Workflow Automation (Rule-Based & GenAI-Driven), Data Cleaning & Validation, Automated Reporting (CSV), Microsoft Excel (Formulas, Formatting, Charts), Dashboarding (Streamlit), Matplotlib, Seaborn

Low-Code & Tools: Microsoft Power Platform, Git, GitHub, Docker

RELEVANT PROJECTS

Website-Based RAG Pipeline & Chatbot

Python | LangChain | Hugging Face Transformers | Qwen2.5 | Streamlit | PyTorch

- Designed and built an end-to-end Retrieval-Augmented Generation (RAG) pipeline that ingests unstructured web text, chunks it (1000-token chunks, 200-token overlap), and indexes vector embeddings for fast, accurate knowledge retrieval.
- Implemented context retrieval using dense vector embeddings and similarity search, surfacing the top-3 most relevant chunks per query for source-grounded responses.
- Used LangChain to orchestrate retrieval and generation, integrating an open-source LLM via Hugging Face Transformers to produce context-restricted answers, with prompt design and iterative testing to constrain outputs strictly to verified source data and reduce hallucination.
- Designed a multi-turn Streamlit interface with a debug panel exposing retrieved context per query, enabling manual validation of AI-generated outputs for accuracy and responsible use before deployment.

Multi-Modal Smart Attendance System

Python | Streamlit | PyTorch | Scikit-learn | OpenCV | Supabase

- Built an end-to-end Python, Streamlit, and Supabase data pipeline that automated 90%+ of manual data entry, quality checks, and reporting tasks, verified across 100+ student records.
- Trained and evaluated dual biometric validation models (Face Recognition + Voice Authentication) using PyTorch and Scikit-learn, achieving 95%+ accuracy and cutting data-quality issues by ~85%.
- Executed a batch data-processing pipeline across 4-5 datasets simultaneously, generating automated CSV reports for 50+ students in under 10 seconds and cutting manual reporting time by ~70%.
- Delivered a QR/link-based enrollment system with bcrypt password hashing and rule-based, role-based access control for secure record management.

EDUCATION

Bachelor of Technology (B.Tech) – Artificial Intelligence & Machine Learning

2023 – 2027

Gyan Ganga Institute of Technology and Sciences, Jabalpur, M.P.

CERTIFICATIONS

Generative AI (IBM) • Python Essentials 1 & 2 (Cisco) • Cyber Security Essentials (Cisco) • Introduction to Modern AI (Cisco) • Data Analytics Essentials (Cisco) • Apply AI and Analyze Customer Reviews (Cisco)

EXTRACURRICULAR ACTIVITIES

- Technical Head** — Gyan Ganga Institute of Technology and Sciences
 - Coordinated end-to-end planning and execution of 5+ college technical events under fixed deadlines, delivering on-time completion and smooth operations across all departments, collaborating closely with student and faculty teams.
- Class Representative (CR)** — Gyan Ganga Institute of Technology and Sciences
 - Represented a cohort of 60+ students as primary liaison between students and faculty, communicating concerns clearly to stakeholders and driving effective resolution of academic and scheduling issues.